

Midterm 1 Part c

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I. QUESTION (10 POINTS)

Write a function

```
def e(N)
```

which calculates e according to the following equation.
Do not assume there is a factorial function defined.

$$e = 1 + \frac{1}{1!} + \frac{1}{2!} + \dots + \frac{1}{N!}$$

II. QUESTION (10 POINTS)

Write a function

```
def containsK(a, K)
```

which checks if the array a contains two numbers whose sum is K . Returns true if there are two such numbers, false otherwise.